

Statistical Scraping
Interest Group

WEB-FOSS-NL

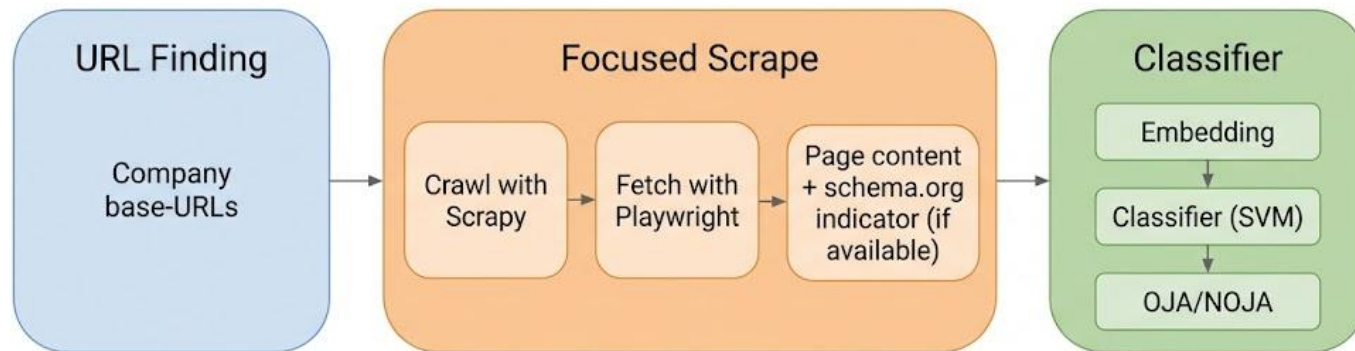
Statistical Scraping explored in the Netherlands

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Statistical Scraping Interest Group meeting 3, 16-17 September in Porto

Summary of the project so far

- Use webscraping to find OJA data for companies in business register
 - Compare findings with our own quarterly survey
- Open source, modular setup for generic statistical scrape



Summary of the project so far

- Improvements to URL Finding
- First test run
 - Using small dataset of ~3000 URLs
 - Classifier pretrained on subset of Dutch WIH data
 - Results promising, especially with model finetuning
- Next steps after SSIG 2 meeting:
 - Explore URL finding with Open Web Index
 - Test run 2: apply URL finding on sample on Job Vacancy Survey population
 - Improving scrape capture and scrape runtime
 - Evaluate and improve classifier on test run 2 output



Recent progress of W3: Focused scrape & Classifier



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Improvements in scraping software

- Use of scrapy library for **async scraping**
 - Plus parallelization = performance boost
- **JS support** by applying playwright for dynamic content
 - Fetch all content, not just HTML skeleton
- Inspired by SSIG2: Look for **schema.org** on web page
 - Look for '**JobPosting**' tag, high confidence indicator?
- Allow (limited) **domain jumps**, with blacklisted domains
 - social media sites and known job portals
- Improved **sitemap** support

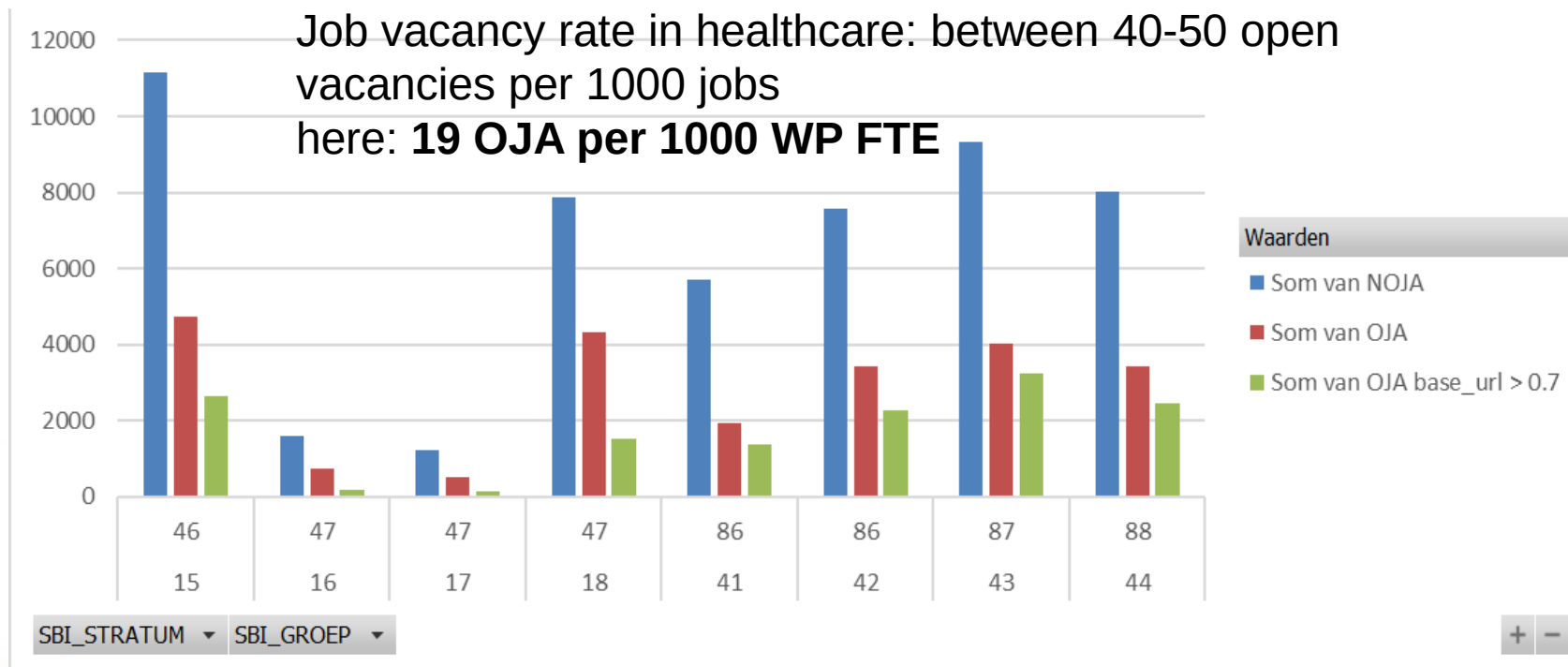


Test run 2: June 30

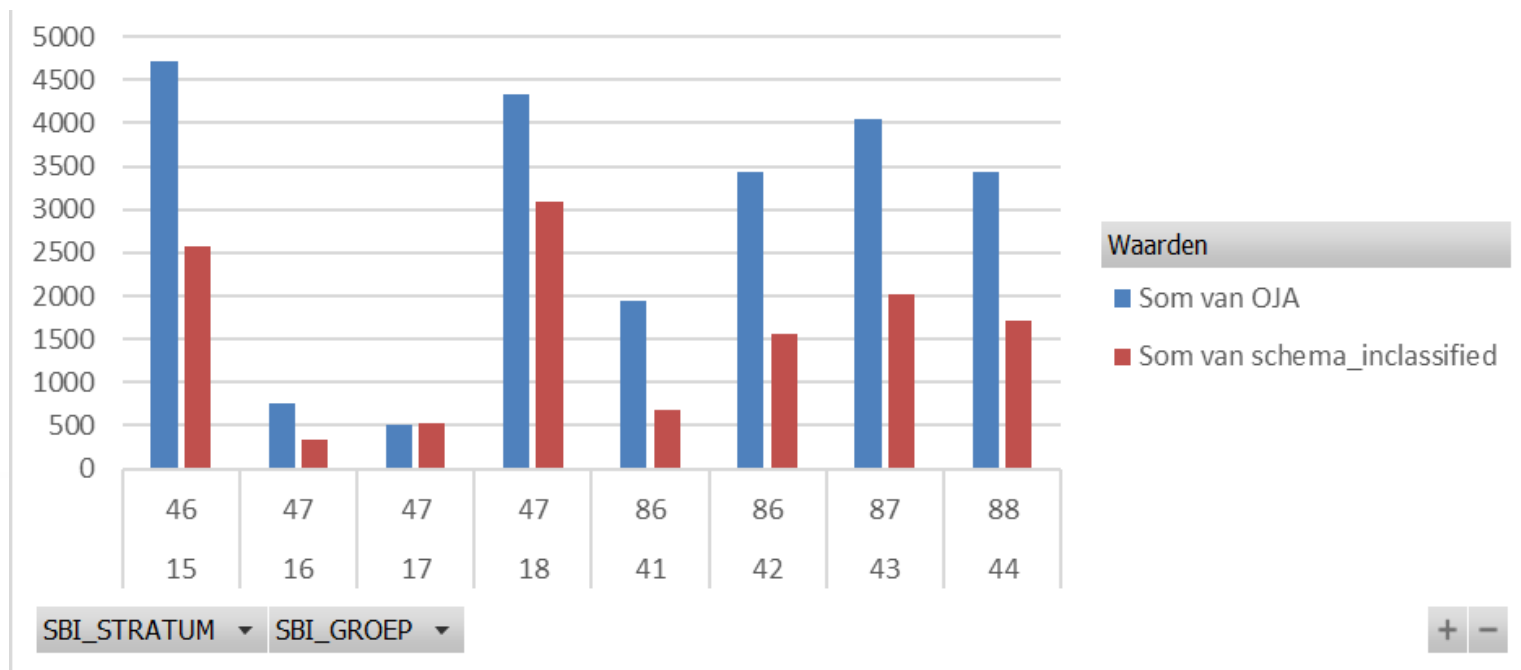
- Effectively ran July 1-14 due to a datacenter malfunction on June 30
- Sample based on the **survey sample of the Job Vacancy Statistic**: All sampled (5933) companies in:
 - Human health and social work activities
 - Wholesale & retail trade (except cars)
- For this selection URL Finding (WP2) provided **4039 unique base-URLs**
- Leading to ~120k pages fetched



OJAs in results of testrun by NACE



OJAs (23k) vs webpages with JobPosting tag (12k)



Overlap classifier and JobPosting tag

OJA_label vs schema_indicator

- Schema_indicator
 - True implies OJA according to website
 - False implies missing indicator/nothing
- Most interesting:
 - schema_indicator: True, classifier: False
 - Why does our classifier disagree?
 - Look at (very) small sample:
 - Mostly FNs by classifier
- Reliable indicator, but often missing
 - Indicator cannot serve as full replacement
 - ~16% of webpages scraped had schema OJA indicator
 - ~50% of found OJAs had schema indicator

OJA_label (predicted)	True	False
	15769	8598
True		
False		
		schema_indicator
		False
		True

Confusion matrix showing the overlap between OJA_label (predicted) and schema_indicator.

OJA_label (predicted)	True	False
True	15769	8598
False	47465	4168



Lessons from testrun 2

- Some base-URLs are **low quality** and heavily influence outcomes
- **Jumping domains** can certainly cause loss of connection with company from base-URL
 - Associate OJAs with from company
- Multiple base-urls **lead to the same domains**, e.g. werkenbijdeoverheid.nl
 - Duplicate download if visited on separate scrapy workers



Lessons from testrun 2

- Current classifier is trained on data from other sources, not our own scrape
- We want more labelled data, especially of **NOJAs that are “close to” an OJA**, in particular:
 - vacancy overview pages,
 - open calls for applications,
 - calls for volunteers,
 - graduation internships
- Add these categories to our labeling



Current work and outlook WP 3



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Scrape

- **Whitelist domain jumps** based on rules
- Improved **dealing with jumps and links**:
 - Reduce duplicate visiting of domains for parallel sessions
 - Attribute OJAs from these sites to all candidate companies e.g. based on company size
- Run test on **September 30** with same company sample as in June
 - Test run with every quarterly survey from now on?



Labeling

- Labeling with three people
 - Always overlap labeling with another person
- Find texts where labelers conflict
 - Discuss / use third as tiebreaker
- Output: labeled dataset with extended labels
- Use full dataset for evaluation first
- After: fine-tune model on subset, validate on rest
 - Measure effect of finetuning
 - Will we need to finetune every time?



Labelling for finetuning OJA classifier

```
Windows PowerShell
URL: https://www.werkenbijvandervalkhotelschiedam.nl/Van-der-Valk-Hotel-Schiedam/vacatures/Front-Office-Shiftleader-JPNLX3CFA.html
Base URL: https://www.hotelheerlen.nl
LABELS: [1] OJA (NL) | [2] OJA (Buitenland) | [3] Stages/Afstuderen | [4] Vrijwilligers | [5] Open sollicitatie | [6] Rest
CONTROL: [7] Stop/Save | [8] Back [9] Remove Label

Content

ROW: 415 | CURRENT LABEL: -1
-----
Solliciteer
Front Office Shiftleader
Van der Valk Hotel Schiedam |
Horeca / Toerisme / Recreatie|
Schiedam |
38 uur
| MBO - HBO
Werken bij Valk Exclusief Hotel Schiedam voelt als tijd doorbrengen met vrienden en familie. Jouw 200 collega's verwelkomen je graag om samen de gasten een
topverblijf te bieden. Door het aantal collega's merk je al dat het een groot hotel is. Dat vind jij kicke, want jij bent iemand die altijd jouw hoofd koel
houdt en de helicopter view bewaart. Jij hebt een duidelijke visie als het gaat om gastvrijheid. Jij bent continue op zoek naar verbetering en streeft naar het
hoogst haalbare. Ook ben jij een voorbeeld in de omgang met allerlei soorten gasten. Ons front office team bestaat uit 12 medewerkers. Als Shiftleader, en
tevens rechterhand van onze Front Office manager, zorg jij ervoor dat de shift goed verloopt. Je traint en motiveert je Front Office collega's om elke dag het
beste uit zichzelf te halen. Door de weeks, in het weekend, met de feestdagen, elke dag leg jij onze gasten graag in de watten! Nieuwsgierig naar onze familie?
Come join the family!
Welke uitdagende taken ga jij allemaal vervullen?
Samen met de Front Office Manager ben jij verantwoordelijk voor de aansturing van het receptieteam en het verloop van de dagelijkse werkzaamheden met als doel
samen tot een topprestatie te komen. Is de Front Office Manager niet aanwezig? Dan ben jij het aanspreekpunt van de front office voor de gasten en medewerkers;
Verantwoordelijk voor het opleiden en inwerken van medewerkers. Jij hebt immers veel ervaring in receptiewerkzaamheden;
Afwikkelen van klachten, complexe reserveringsaanvragen, adhoc situaties en calamiteiten;
Mede verantwoordelijk voor het verhogen en continueren van het serviceniveau;
Verwelkomen, te woord staan, in- en uitchecken van (binnen- en buitenlandse) gasten;
Toezien op correcte omgang met gasten/relaties;
Uitvoeren verschillende (ondersteunde) administratieve werkzaamheden;
Gastvrijheid betekent voor jou net die stap extra zetten om ervoor te zorgen dat onze gasten het hotel verlaten met een glimlach.
Functievereisten
Je beschikt over een MBO/HBO werk- en denkniveau;
Minimaal 1 á 2 jaar ervaring in een soortgelijke functie;

Progress: 416/10000 | Current Label: -1 | NaNs in content: 0 | #Rows labeled: 415 | Label distribution: [9585, 214, 151, 16, 15, 11, 8]
```

Reconsider prompting?

- LLMs keep getting better, faster
 - Internal LLM support is getting better
 - Can add information from business registry to prompt without leakage
- A good prompt will likely perform similar to current classifier
 - Without labeling, training, finetuning..
 - Still need to validate!

oI qwen3-8-27b

Is the following text OJA? make no mistakes



qwen3-8-27b



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Reconsider prompting?

- Scaling issue remains!
 - If 1 prompt takes ~30s, 100K texts = ~34 days
 - Embedding + Classifier takes ~ 1 day
- Cascading classifier?
 - Filter with cheaper methods before moving to expensive methods
 - First rule based approach (e.g. schema indicator + keyword matching)
 - Not enough? → Embedding + SVM predicts OJA probability
 - Too close to 50%? → Prompt LLM



Next steps

- Finish labeling, evaluate results
- Work towards next release focused scrape
 - Improvements in scrape/performance
 - Improve parallelisation, coding agent promises ~5-10x speedup, will need to verify
- Keep simultaneous run with survey
 - Compare trends in scrape/survey
- Compare Classifier, Schema, and Survey OJA labeling for specific companies



OWI experiment



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URL finding: OWS / OWI experiment

- Goal: explore the value of OWI data to enhance Dutch SBR
- OWI datasets: <https://openwebindex.eu/datasets>
- Downloaded data marked as Dutch (1.5M records, 41.2 GB)
*owilix remote pull all/id=<ID> --files==**/language=nld/**
- Removed non-Dutch (mostly .be) and without address or phone_nr => 18K unique domains: .nl .eu .net .org
- 17K could easily be linked to SBR on zip, house_nr, phone_nr
- Nick: “the total correctness has not yet been verified, but the ~100 sampled entries looked promising”
- Note1: working with such large dataset internally was a challenge
- Note2: OWI data contains over 60 variables, only few explored

